

Mastering Averages (Quantitative Aptitude)

Comprehensive Study Guide, 15 Solved Examples, and 25 Practice Questions with Answer Key

1. Comprehensive Theory of Averages

What is an Average?

The average (or arithmetic mean) of a given group of values is a single value that represents the central tendency of the entire group. It is calculated by dividing the sum of all observations by the total number of observations.

$$\text{Average} = (\text{Sum of Observations}) / (\text{Total Number of Observations})$$

$$\text{Sum of Observations} = \text{Average} \times \text{Total Number of Observations}$$

Key Properties & Shortcuts for Competitive Exams:

- If each observation is increased by a constant k , the new average also increases by k .
- If each observation is multiplied by k , the new average is also multiplied by k .
- **Combined Average:** If a group of n_1 items has an average A_1 and another group of n_2 items has an average A_2 , then:

$$\text{Combined Avg} = (n_1 A_1 + n_2 A_2) / (n_1 + n_2)$$

- **Consecutive Numbers / Arithmetic Progression (AP):** For any set of consecutive numbers (or numbers in AP), the average is exactly the middle number. If the count is even, the average is the mean of the two middle numbers. Also, $\text{Average} = (\text{First Term} + \text{Last Term}) / 2$.

2. Solved Examples (15 Detailed Solutions)

Example 1: Basic Average Calculation

Find the average of the first 5 prime numbers.

Solution: First 5 prime numbers are 2, 3, 5, 7, and 11.

$$\text{Sum} = 2 + 3 + 5 + 7 + 11 = 28.$$

$$\text{Total numbers} = 5.$$

$$\text{Average} = 28 / 5 = 5.6$$

Example 2: Average of Consecutive Numbers

Find the average of all odd numbers from 1 to 49.

Solution: The odd numbers form an arithmetic progression: 1, 3, 5, ..., 49.

Formula for AP average: $(\text{First Term} + \text{Last Term}) / 2$

$$\text{Average} = (1 + 49) / 2 = 50 / 2 = 25$$

Example 3: Finding Missing Value

The average weight of 4 boys is 50 kg. If three of them weigh 45 kg, 52 kg, and 48 kg respectively, find the weight of the fourth boy.

Solution: Total weight of 4 boys = $4 \times 50 = 200$ kg.

Sum of weights of 3 boys = $45 + 52 + 48 = 145$ kg.

Weight of 4th boy = $200 - 145 = 55$ kg.

Example 4: Addition of New Member

The average age of 10 students is 15 years. If the teacher's age is included, the average age increases by 1 year. Find the age of the teacher.

Solution: Total age of 10 students = $10 \times 15 = 150$ years.

Total age of 10 students + teacher (11 persons) = $11 \times (15 + 1) = 11 \times 16 = 176$ years.

Teacher's age = $176 - 150 = 26$ years.

Example 5: Replacement Concept

The average weight of 8 persons increases by 2.5 kg when a new person comes in place of one weighing 65 kg. Find the weight of the new person.

Solution: Total weight increased = $8 \times 2.5 = 20$ kg.

Weight of new person = Weight of replaced person + Total increase

Weight = $65 + 20 = 85$ kg.

Example 6: Combined Average

A class has 30 boys whose average marks are 60, and 20 girls whose average marks are 70. Find the combined average marks of the whole class.

Solution: Total marks of boys = $30 \times 60 = 1800$.

Total marks of girls = $20 \times 70 = 1400$.

Combined Average = $(1800 + 1400) / (30 + 20) = 3200 / 50 = 64$.

Example 7: Batting Average / Cricket Runs

A batsman has a certain average for 11 innings. In the 12th inning, he scores 90 runs, thereby decreasing his average by 2 runs. Find his average after 12 innings.

Solution: Let the average after 12 innings be x .

Average after 11 innings = $x + 2$.

Total runs in 12 innings = $12x$. Total runs in 11 innings = $11(x + 2)$.

$12x - 11(x + 2) = 90 \Rightarrow 12x - 11x - 22 = 90 \Rightarrow x = 90 + 22 = 112$.

Example 8: Temperature / Weather Problem

The average temperature of Monday, Tuesday, and Wednesday was 38°C . The average temperature of Tuesday, Wednesday, and Thursday was 40°C . If the temperature of Thursday was 42°C , find the temperature on Monday.

Solution: $\text{Mon} + \text{Tue} + \text{Wed} = 38 \times 3 = 114^{\circ}\text{C}$ — (1)
 $\text{Tue} + \text{Wed} + \text{Thu} = 40 \times 3 = 120^{\circ}\text{C}$ — (2)
Subtracting (1) from (2): $\text{Thu} - \text{Mon} = 120 - 114 = 6^{\circ}\text{C}$.
Since $\text{Thu} = 42^{\circ}\text{C}$, $\text{Mon} = 42 - 6 = \mathbf{36^{\circ}\text{C}}$.

Example 9: Speed and Average Speed

A car travels from city A to city B at a speed of 40 km/h and returns from B to A at a speed of 60 km/h. Find the average speed for the entire journey.

Solution: When distance is the same, Average Speed = $\frac{2xy}{x+y}$.
Average Speed = $(2 \times 40 \times 60) / (40 + 60) = 4800 / 100 = \mathbf{48 \text{ km/h}}$.

Example 10: Age Group Problem

The average age of a family of 4 members is 28 years. If the age of a baby born recently is included, what is the new average age of the family?

Solution: Total age of 4 members = $4 \times 28 = 112$ years.
Total family members now = 5 (including baby whose age = 0).
New Average = $112 / 5 = \mathbf{22.4 \text{ years}}$.

Example 11: Consecutive Even Numbers

The average of 5 consecutive even numbers is 42. Find the largest number in the set.

Solution: For consecutive numbers, the average is always the middle number.
Middle number (3rd number) = 42.
Since they are even numbers, the 5 numbers are 38, 40, 42, 44, 46.
The largest number is **46**.

Example 12: Misread Observations

The average of 20 numbers was calculated as 35. It was later discovered that a number 56 was misread as 36. Find the correct average.

Solution: Incorrect sum = $20 \times 35 = 700$.
Correct sum = Incorrect sum – Incorrect number + Correct number = $700 - 36 + 56 = 720$.
Correct average = $720 / 20 = \mathbf{36}$.

Example 13: Expenditure and Savings

A man's average monthly expenditure for the first 4 months of a year is \$1,200 and for the next 8 months is \$1,500. If he saves \$4,200 that year, find his total annual income.

Solution: Expenditure for first 4 months = $4 \times 1200 = \$4,800$.

Expenditure for next 8 months = $8 \times 1500 = \$12,000$.

Total annual expenditure = $4800 + 12000 = \$16,800$.

Total Annual Income = Expenditure + Savings = $16800 + 4200 = \$21,000$.

Example 14: Group Exclusion

The average weight of 25 students is 50 kg. If the teacher's weight is excluded, the average weight reduces by 1 kg. Find the weight of the teacher.

Solution: Total weight of 25 students + teacher (26 persons) = $26 \times 50 = 1300$ kg.

Total weight of 25 students alone (excluding teacher) = $25 \times (50 - 1) = 25 \times 49 = 1225$ kg.

Teacher's weight = $1300 - 1225 = 75$ kg.

Example 15: Multiples Average

Find the average of the first 10 multiples of 7.

Solution: First 10 multiples of 7 are 7, 14, 21, ..., 70 (which form an AP).

Average = $(\text{First Term} + \text{Last Term}) / 2 = (7 + 70) / 2 = 77 / 2 = 38.5$.

3. Practice Questions (25 MCQs)

Q1. Find the average of first 20 natural numbers.

(A) 10

(B) 10.5

(C) 11

(D) 11.5

Q2. The average of 5 numbers is 27. If one number is excluded, the average becomes 25. Find the excluded number.

(A) 25

(B) 27

(C) 35

(D) 40

Q3. What is the average of all prime numbers between 1 and 20?

(A) 9.25

(B) 9.59

(C) 10.12

(D) 10.5

Q4. The average of 4 consecutive odd numbers is 24. Find the smallest number.

(A) 21

(B) 23

(C) 19

(D) 17

Q5. A batsman makes a score of 87 runs in the 17th inning and thus increases his average by 3. Find his average after 17th inning.

(A) 35

(B) 38

(C) 39

(D) 42

Q6. The average age of husband and wife was 23 years when they were married 5 years ago. Now the average age of family (husband, wife, and child) is 20 years. Find the age of the child.

- (A) 1 year (B) 2 years (C) 3 years (D) 4 years

Q7. The average of 10 numbers is 40. Later it was found that number 65 was misread as 45. Find the correct average.

- (A) 41 (B) 42 (C) 41.5 (D) 42.5

Q8. The average weight of 10 containers is 56 kg. If the weight of 3 containers is 50 kg each and the weight of 4 containers is 60 kg each, find the weight of the remaining 3 containers (assuming they are equal).

- (A) 52 kg (B) 54 kg (C) 55 kg (D) 56 kg

Q9. Find the average of the squares of the first 5 natural numbers.

- (A) 10 (B) 11 (C) 12 (D) 13

Q10. The average of 6 numbers is 12. If each number is multiplied by 3, what is the new average?

- (A) 12 (B) 24 (C) 36 (D) 72

Q11. The mean of 50 observations was 36. It was found later that an observation 48 was wrongly taken as 84. The corrected mean is:

- (A) 35.28 (B) 36.12 (C) 34.50 (D) 35.80

Q12. The average monthly income of P and Q is \$5,050. The average monthly income of Q and R is \$6,250 and that of P and R is \$5,200. Find the income of P.

- (A) \$4,000 (B) \$4,500 (C) \$5,000 (D) \$5,500

Q13. In a school with 600 students, the average age of boys is 12 years and that of girls is 11 years. If the total average age is 11 years 9 months, find the number of girls.

- (A) 150 (B) 200 (C) 250 (D) 300

Q14. The average of 7 consecutive numbers is 20. The largest number of this set is:

- (A) 20 (B) 22 (C) 23 (D) 24

Q15. Five years ago, the average age of A and B was 20 years. With C now joining them, the average age becomes 25 years. How old is C now?

- (A) 24 years (B) 25 years (C) 27 years (D) 30 years

Q16. The average weight of a class of 24 students is 35 kg. If the weight of the teacher is also included, the average weight increases by 400 g. What is the weight of the teacher?

- (A) 45 kg (B) 48 kg (C) 50 kg (D) 52 kg

Q17. A man covers a distance of 50 km at 10 km/h, another 50 km at 25 km/h, and 50 km at 50 km/h. Find his average speed for the entire journey.

- (A) 17.65 km/h (B) 18.75 km/h (C) 20.00 km/h (D) 22.50 km/h

Q18. The average of 8 observations is 12. If the average of the first 5 observations is 10 and the average of the last 4 observations is 15, find the 5th observation.

- (A) 18 (B) 20 (C) 22 (D) 24

Q19. The sum of 5 numbers is 555. The average of the first two numbers is 75 and the third number is 115. Find the average of the last two numbers.

- (A) 145 (B) 150 (C) 155 (D) 160

Q20. The average of three numbers is 135. The largest number is 195, and the difference between the other two is 20. Find the smallest number.

- (A) 80 (B) 90 (C) 95 (D) 100

Q21. If the average of x , y , and z is equal to the average of p , q , and r , which of the following is true if $x+y+z = p+q+r$?

- (A) Always true (B) Verified (C) Consistent (D) All equal sums

Q22. A library has an average of 510 visitors on Sundays and 240 on other days. The average number of visitors per day in a month of 30 days beginning with Sunday is:

- (A) 275 (B) 285 (C) 295 (D) 305

Q23. The average of first 9 whole numbers is:

- (A) 4 (B) 4.5 (C) 5 (D) 5.5

Q24. Three years ago, the average age of a family of 5 members was 17 years. A baby having been born, the current average age of the family is the same today. The present age of the baby is:

- (A) 1 year (B) 1.5 years (C) 2 years (D) 3 years

Q25. The average of 11 numbers is 50. If the average of the first 6 numbers is 49 and that of the last 6 numbers is 52, find the 6th number.

- (A) 52 (B) 54 (C) 56 (D) 58

4. Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	B (10.5)	6	C (3 yrs)	11	A (35.28)	16	B (48 kg)	21	A
2	C (35)	7	A (41)	12	A (\$4k)	17	B (18.75)	22	B (285)
3	B (9.59)	8	B (54 kg)	13	A (150)	18	A (18)	23	A (4)
4	A (21)	9	D (13)	14	C (23)	19	C (155)	24	A (1 yr)
5	C (39)	10	C (36)	15	B (25 yrs)	26*	C (56)		

*Note: Q25 corresponds to answer C (56).